Laiba Mazhar

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EDUCATION

FAST National University of Computer and Emerging Sciences Aug 2022 – May 2026
*BS Data Science | Islamabad, Pakistan | **CGPA 3.12 / 4.00***

- Final year project: **NexusPilot** — an integrated business platform unifying finance, HR and CRM workflows over a 15-entity relational schema.
- Relevant coursework: Machine Learning, Data Mining, Artificial Intelligence, Parallel and Distributed Computing, Database Systems, Statistics, Data Warehousing, Computer Networks.
- **Higher Secondary (FSc Pre-Engineering)**, Aspire College, 2019–2021 — 1024/1100 (93.09%), 2nd position in college, fully funded scholarship.
- **Secondary (Matriculation)**, Government Girls High School, Umer Block, Lahore, 2017–2019 — 1010/1100 (91.82%), 1st position in class, fully funded scholarship.

RESEARCH

Four research manuscripts. None is currently published; status is stated explicitly for each.

Anchored Counterfactual Attribution of Model Degradation to ETL Pipeline Operators 2026
***First author**, with M. Sarfraz | Manuscript in preparation | github.com/laiba-mazhar/culpa-pipeline-attribution*

- **Abstract.** When a production model degrades between two scheduled pipeline runs with no code deployed, engineers must determine which operator is responsible. We formulate this as a cooperative game over *hybrid replays* of a fixed pipeline across two temporal snapshots, whose Shapley value decomposes the observed degradation exactly across operators. We show the standard Shapley value is correct as accounting but poor as triage, and introduce the **incident-anchored Shapley value**, which recovers both standard Shapley and leave-one-out ablation as boundary cases. On real NYC taxi partitions it reaches 100% top-1 accuracy where leave-one-out drops to 50%, while retaining exact decomposition to 3×10^{-17}. Output-hash pruning reduces 2^{33} hybrid replays to three model fits.
- Open artifact with eight reproducible experiments, full run logs and an IEEE-format manuscript. Also establishes that current stagewise practice is order-dependent: attributed blame reverses sign across valid topological orders.

Adversarial Data Poisoning Detection in Federated Learning via ICS 2025
Sole author | Manuscript in preparation

- **Abstract.** Proposes the *Independent Class-wise Coherence Score* (ICS), a defense against label-flipping and backdoor attacks in federated learning, using MI-FGSM adversarial probing and four fused coherence metrics. Across 15 distributed clients on MNIST, EMNIST and CIFAR-10, ICS outperforms global-coherence baselines by 5–7% accuracy and reduces attack success rate by 30–40% (75.02% accuracy at 19.75% ASR on MNIST).

Robust and Explainable Multiclass Intrusion Detection with XGBoost and SHAP 2025
Sole author | Manuscript in preparation

- **Abstract.** A statistically validated, explainable multiclass intrusion detection system combining XGBoost, TreeSHAP and a severity scoring engine for SOC deployment, evaluated on NSL-KDD and UNSW-NB15. Achieves 99.90% average accuracy, a 12.28% macro-F1 gain over a Random Forest baseline and a 2.44× stability improvement across five seeds, emitting JSON-exportable SOC alert packets.

LLM-Driven Autonomous Trading Agents with Hallucination Mitigation 2025
Sole author | Manuscript in preparation

- **Abstract.** A three-agent pipeline (Analyst, Critic, Decision) built on Llama-3.3-70B to mitigate hallucination in financial reasoning over real-time market data. The Critic agent emits structured JSON verification reports, achieving an 80% recommendation adjustment rate and consistent hallucination detection across five major US equities.

WORK EXPERIENCE

CloudWorks — Texas, USA (Remote) Jun 2026 – Present
Data Developer and Analyst | Part-time

- Build and maintain data pipelines and analytics for US healthcare clients, including CareFlow, a care-coordination platform with role-based access control, HIPAA-compliant data handling and full audit logging.

Al Muttaqeen Institute — Remote Jan 2026 – Jun 2026
Database Manager

- Built automated validation, reporting and reconciliation pipelines for institutional records, improving cross-departmental data accuracy and audit readiness.

Tashi Technologies Corp — Islamabad, Pakistan Jul 2025 – Oct 2025

Data Automation Engineering Intern

- Built ETL workflows and classification/regression pipelines (XGBoost, scikit-learn); integrated AI APIs for process automation and deployed models to cloud infrastructure for real-time scoring.

Independent / Freelance — Remote Dec 2024 – Present

Data and AI Engineer, international clients

- Delivered batch and streaming pipelines (Spark, Kafka, Airflow) on GCP and AWS feeding ML models and BI dashboards; built LLM prototypes with RAG and agentic workflows.

SELECTED TECHNICAL PROJECTS

- **English–Urdu Neural Machine Translation** — Transformer built from scratch with an LSTM/Bahdanau-attention baseline, BPE tokenization and BLEU/ROUGE evaluation.
- **ragforge** — hybrid retrieval (dense + BM25 + reciprocal rank fusion) with a built-in evaluation harness; NumPy-only core.
- **Real-Time Telecom Network Analytics** — Kafka/Spark/Airflow streaming pipeline with CI/CD and DAG version control via GitHub Actions.
- **LaunchMind** — multi-agent system that autonomously operates a micro-startup workflow.

TECHNICAL SKILLS

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, federated learning, adversarial ML, explainable AI (SHAP), NLP, time-series forecasting, LLM agents, RAG.
- **Data engineering:** Apache Spark, PySpark, Kafka, Airflow, dbt, Databricks, Snowflake, BigQuery, ETL/ELT design.
- **Programming:** Python, SQL, TypeScript/JavaScript, C/C++, Java (basic), MASM assembly.
- **Infrastructure:** GCP, AWS, Docker, GitHub Actions, PostgreSQL, MongoDB, Supabase, FastAPI, Django REST.
- **Analysis and visualization:** Power BI, Tableau, Matplotlib, Seaborn, LaTeX.

AWARDS AND CERTIFICATIONS

- **Winner, Big Data Quest Hackathon** — DataFest-25, NaSCon: predictive analytics solution built on PySpark and cloud infrastructure.
- **Speed Programming Certificate** — NaSCon-24.
- **Machine Learning**, Stanford University (Coursera, Andrew Ng), 2026.
- **Certificates of Special Mention**, Model United Nations — CUIMUN (COMSATS), 2023; BUIMUN (Islamabad), 2024.
- **Winner**, university arts competition — FAST-NUCES.
- Academic distinction shields at secondary and higher-secondary level.

EXTRACURRICULAR ACTIVITIES AND INTERESTS

Rah-e-Haq (NGO) — Islamabad Nov 2023 – Apr 2025

Donation Officer, volunteer

- Maintained structured donation and financial records; coordinated cross-team deliverables and transaction accountability.

Baitulnoor — Pakistan 1.5 years

Head of Investigations, volunteer

- Led verification and investigation processes across cases; documented findings into actionable stakeholder reports.

- **Founder, Datahelyx** — a data and AI consultancy (datahelyx.com), currently building its first client engagements.
- **Founder and artist, Rangrayze** — a gallery for Sufi-inspired calligraphy and canvas work; manages creative production, commissions and digital marketing.
- **Founder, airRTH** — a community climate initiative running sustainability awareness and environmental action programs, at an early stage.
- **Model United Nations** — delegate across multiple national conferences; two Special Mention awards.
- **Sport** — three medals in inter-college competition.
- **Interests** — trustworthy and diagnosable ML systems, data engineering for AI, distributed systems, calligraphy, climate advocacy, competitive debate.

LANGUAGES

English (fluent, professional working proficiency) | Urdu (native) | German (basic) | Arabic (basic reading and writing)